Summary Report:

The overall task involved segmenting and identifying white mineral patches and gray coal patches from an input image. This task was accomplished through a series of image processing techniques implemented in Python using the OpenCV library.

1. **Preprocessing**:
    - The input image was loaded and converted to grayscale.
    - Fixed thresholds were initially used to separate the light and dark gray pixels, with the light gray pixels representing potential mineral patches.
    - To handle varying brightness levels, adaptive thresholding was employed for segmenting the coal patches, which calculates local thresholds based on the neighborhood of each pixel.
2. **Segmentation**:
    - For mineral patches, the light gray pixels were set to white (255), and the dark gray pixels were set to black (0), creating a binary image with the mineral patches highlighted in white.
    - For coal patches, a binary mask was created using adaptive thresholding, separating the coal patches from the background.
    - Morphological operations were applied to clean up the binary masks, removing noise and filling small holes.
3. **Contour Detection and Filtering**:
    - Contours were detected on the binary masks, representing the regions of interest (mineral patches or coal patches).
    - A minimum area threshold was applied to filter out small contours that may not represent actual coal patches.
4. **Bounding Boxes and Visualization**:
    - Bounding rectangles were calculated for the remaining contours, enclosing the segmented mineral patches and coal patches.
    - These bounding boxes were drawn on a copy of the original image for visualization purposes.
5. **Bounding Box Merging for Coal Patches**:
    - To handle overlapping coal patches, a function was implemented to merge bounding boxes based on their overlapping area.
    - This function took the stitched image dimensions and a threshold for the overlap ratio to determine if bounding boxes should be merged.
    - The merged bounding boxes were drawn on the stitched image, providing a better representation of the coal patches.
6. **Output**:
    - The processed images with bounding boxes around the segmented mineral patches and coal patches were displayed for visual inspection.
    - Additionally, the code provides the flexibility to save the processed images with bounding boxes to disk for further analysis or archiving purposes.

The implemented solution leveraged various image processing techniques, including thresholding, morphological operations, contour detection, bounding box calculation, and bounding box merging. The code was designed to handle varying brightness levels and overlapping regions, making it more robust and adaptable to different input images.

Overall, the tasks performed successfully segmented and identified the white mineral patches and gray coal patches from the input image, providing a foundation for further analysis, quantification, or classification of these regions of interest.